Generalization-Aware Zero-Shot Neural Architecture Search for Self-Supervised Transformers

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***Abstract*—Neural Architecture Search (NAS) aims to automate the design of neural networks, enabling the discovery of highly effective architectures. Recent advancements in NAS have shown significant success in identifying high-performing Transformer architectures for computer vision and natural language processing tasks. However, most NAS research has focused on supervised learning frameworks, which rely heavily on labeled data. This dependence on labeled data makes deploying these methods in real-world applications challenging due to the high cost of data annotation. Additionally, previous studies often prioritize model performance while neglecting generalization ability, particularly in scenarios with limited labeled data. To address these challenges, this study introduces a generalization-aware zero-shot proxy based on self-supervised learning. By combining this proxy with a complementary zero-shot proxy, we identify architectures that balance generalization ability and expressivity. Experimental results demonstrate that the architectures discovered using the proposed approach achieve competitive performance on the ImageNet and Wikitext-2 datasets while significantly reducing the required labeled data by up to 75% and 99%, respectively.**

***Index Terms*—Neural Architecture Search, Self-Supervised Learning, Computer Vision, Natural Language Processing**

I. Introduction

Transformer networks [1] have attracted significant attention in recent years due to their effectiveness in processing long-range dependencies within data and their superior performance across a wide range of tasks such as Computer Vision (CV) and Natural Language Processing (NLP). Despite their empirical success, Transformer architectures rely heavily on manual design by human experts, requiring extensive trial-and-error experiments to determine optimal design choices.

Neural Architecture Search (NAS) [2] has emerged as a promising approach to automate the design of neural networks. Among various NAS methods, zero-shot approaches [3] [4] have gained substantial attention. These methods employ zero-shot proxies to estimate a network's performance at the initialization stage, enabling rapid evaluation within seconds by computing statistics derived from a single forward or backward propagation pass using a mini-batch of data [5]. By leveraging automated network design, such methods can uncover custom architectures that match or exceed the performance of manually designed Transformers in CV and NLP tasks. However, most recent NAS research has been conducted within fully supervised learning paradigms, where optimal Transformer architectures are trained using both input data and their corresponding labels. This reliance on labeled data introduces significant limitations, as it substantially increases the cost and effort associated with data annotation, thereby restricting the practical applicability of current NAS techniques. For instance, the annotation of the widely-used ImageNet [6] dataset required the efforts of 49,000 Amazon Mechanical Turk workers for three years. Furthermore, domain expertise is essential for specialized tasks such as medical imaging, requiring professionals like radiologists to accurately label datasets. These challenges underscore the urgent need for NAS methods that can reduce or eliminate dependence on large quantities of labeled data, paving the way for more scalable and cost-effective solutions. As a potential solution, self-supervised learning (SSL), which derives supervisory signals directly from the input data itself, has demonstrated considerable success in mitigating the dependence on labeled data [7] [8]. Despite its promise, there has been limited research on integrating zero-shot NAS with SSL.

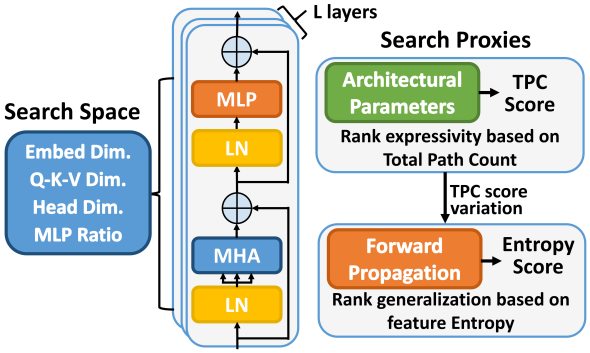

Fig. 1. Overview of the search space and search proxies in the proposed GEN-TPC-NAS.

In this paper, we propose GEN-TPC-NAS, a generalization-aware Entropy and TPC based zero-shot NAS. Figure 1 is the overview of GEN-TPC-NAS. First, we utilize the TPC score [9], an efficient and fast zero-shot proxy that assesses network expressivity [10] based on the concept of Total Path Count (TPC). Second, we introduce the Entropy score, a zero-shot proxy based on the concept of SSL. This proxy evaluates the generalization ability of neural networks by analyzing the eigenvalues corresponding to feature matrixs, which reflect

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feature diversity. By optimizing the search strategy, we combine these two proxies to identify efficient neural architectures that balance expressivity and generalization ability. Finally, we pre-train the searched architectures in SSL manner and fine-tune the models on low-labeled data scenarios to evaluate the effectiveness. The contributions of this paper are summarized as follows:

- We propose the Entropy score, a zero-shot proxy inspired by SSL concept, to measure the generalization ability of neural networks.
- We propose a search algorithm that combines the TPC and Entropy scores to identify architectures balancing expressivity and generalization ability.
- To the best of our knowledge, this is the first work to integrate zero-shot NAS with SSL while explicitly considering network generalization ability. On the ImageNet-1k and Wikitext-2 datasets, our method maintains baseline-level performance while reducing the need for labeled data by up to 75% and 99%, respectively.

II. Related Work

A. Neural Architecture Search

The earliest neural architecture search algorithms adopt Reinforcement Learning [11] that evaluates the performance of the models obtained through training. A complete search process typically requires training hundreds or even thousands of architectures, resulting in overall search times reaching several thousand hours or more, which is extremely resource-intensive and economically inefficient.

To address this problem, one-shot NAS [12] [13] uses the concept of parameter sharing by training a large network. During training, one-shot NAS selects and trains only one subnet at a time from the large network. After training, the weight parameters are placed back into the large network. Although one-shot NAS reduces training computational load and memory usage, the shared weight parameters among subnets can cause mutual interference, making it challenging to ensure that each subnet is fully trained, which can affect model performance.

Zero-Shot NAS [3] [14] [15] [16] does not involve any training of architectures. Instead, it uses a performance predictor to estimate the quality of the searched architectures, significantly reducing search time and computational load. However, earlier zero-shot NAS approaches face several issues. Firstly, many zero-shot NAS performance predictors are typically accurate only for a specific type of neural network [16] [17]. Furthermore, many studies are applied to single tasks like image classification and do not achieve cross-model or cross-application generalization. Lastly, most zero-shot NAS methods focus on the supervised learning framework, limiting the potential to deploy these methods in real applications.

B. Self-Supervised Learning

Self-supervised learning (SSL) uses unlabeled data to learn meaningful representations. SSL relies on pretext tasks that create pseudo-labels from the data itself, helping models generalize effectively to various downstream tasks. GPT-2 [18] uses an autoregressive language modeling objective, predicting the next token in a sequence based on the context. By training on large text datasets, GPT-2 achieves excellent results on various NLP tasks after fine-tuning on small labeled datasets.

With the development of vision transformers (ViTs) [19] [20] [21], Masked Image Modeling (MIM) gradually replaces the dominant position of contrastive learning [22] [23] in visual self-supervised representation learning due to its superior fine-tuning performance in various visual downstream tasks. MAE [8] masks patches of an input image and trains a vision transformer to reconstruct the missing pixel values. MCMAE [24] proposes a hybrid convolution-transformer to learn more discriminative representations via the mask auto-encoding scheme. MaskTAS [25] introduces a MIM-based self-supervised one-shot NAS framework incorporating knowledge distillation. However, the proposed method is not evaluated in limited labeled data scenarios.

III. Methodology

A. Entropy Score

Generalization ability [26] refers to generalizing from the training data to limited unseen data. A network with strong generalization capability should achieve comparable performance using fewer labeled data, indicating that it has learned meaningful and transferable representations. Given the random initialized weights and inputs, we evaluate the generalization ability of a self-supervised network by the uniformity of feature space. The isotropy of the feature space [27] refers to how uniformly a model's features are distributed across different directions. Studies have shown that networks with higher isotropy at initialization tend to demonstrate better generalization ability [28]. To measure the isotropy of the feature space, we analyze the eigenvalues of the feature matrix. Let $f_k \in R^{\mathrm{c}\times\mathrm{n}}$ be the self-attention feature matrix of the p-th layer in a Transformer with H number of heads, where c is the number of channels, n is the number of tokens. We use singular value decomposition and square each singular value to obtain c eigenvalues λ_1 to λ_c of the feature matrix. If the different orientations in the feature space are distributed more isotropically, the magnitudes of the c eigenvalues become more similar. Consequently, we use the entropy of the c eigenvalues as a metric to evaluate the model's generalization ability:

$$E_h = \sum_{i=1}^{c} -\bar{\lambda}_i \log\left(\bar{\lambda}_i\right), \tag{1}$$

where $\bar{\lambda}_i$ is the normalized eigenvalue. The overall Entropy score is defined by summing up the average entropy of different heads for each layer:

$$S_e = \sum_{p} \frac{1}{H} \sum_{h} E_h, \tag{2}$$

To examine the effectiveness of the proposed Entropy score, we selected 32 ViT models from [4] [20] [21] [24] and conducted MIM pre-training followed by fine-tuning on

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ImageNet-100 low-labeled data. Figure 2 illustrates the relationship between the Entropy score and the percentage of label reduction each model can achieve while maintaining a fixed test accuracy. The results shown in Figure 2 and Table I demonstrate a Spearman's rank correlation coefficient of 0.93, indicating a strong positive correlation between the Entropy score and the generalization ability of the models.

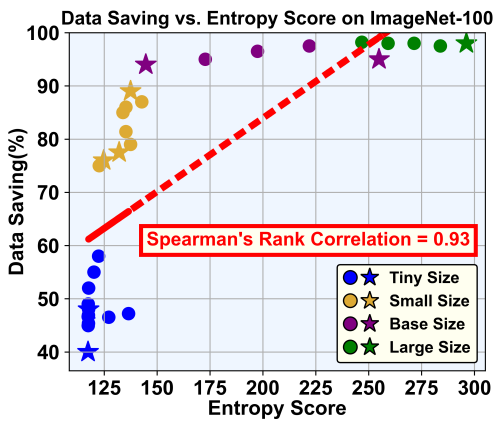

Fig. 2. Relationship between the Entropy score and percentage of label reduction along with Spearman's rank correlation coefficient. The blue, yellow, purple, and green points represent the Tiny, Small, Base, and Large sizes of ViTs. The star markers indicate the models presented in Table I. For this analysis, we set the fixed test accuracy to 80%.

TABLE I
RELATIONSHIP BETWEEN ENTROPY SCORES AND LABELED DATA SAVING FOR DIFFERENT MODELS ON IMAGENET-100.

Model	Entropy Score	Labeled Data Saving (%)
DeiT-T	117.4	40.0
Autoformer-T	117.6	48.0
ResNet-50	125.0	76.0
DeiT-S	132.1	77.5
MCMAE-S	137.5	89.0
DeiT-B	144.7	94.0
Swin-B	254.9	95.0
DeiT-L	296.2	98.0

B. TPC Score

TPC-NAS [9] is a fast zero-shot NAS method. It evaluates an architecture's quality using expressivity [10]. A model with higher architectural expressivity is expected to perform better on practical tasks or applications. Consider a neuron in a network as a node, where multiple paths connect it to nodes in the next layer, each associated with a different weight. A greater number of paths in a model can enable more diverse transformations of the input data. As a result, architectures with more paths generally exhibit better expressivity [10]. For instance, the model in Figure 3 has a total end-to-end path count of 81.

The core of TPC-NAS is the TPC score [9], which calculates the total number of paths from the input layer to the output layer. For a multi-layer model, the TPC score is obtained by

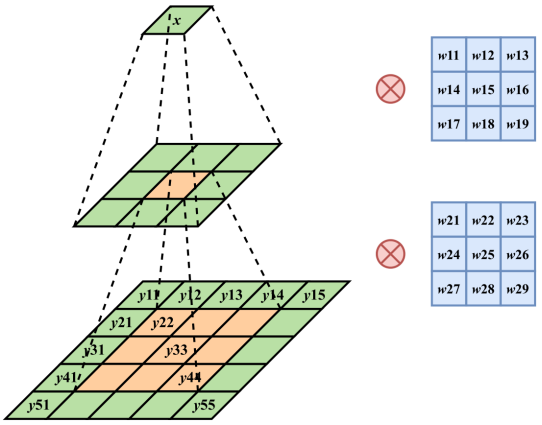

Fig. 3. Example of a model with two convolution layers adapted from [9]. Each input pixel can influence 9 output pixels. Consequently, the entire model has a total of 81 paths.

multiplying the number of paths at each layer, equivalent to the network's outdegree. Equation (3) represents the TPC score for a convolutional layer, where $c_{o,p}$, k_p, s_p, and g_p denote the number of output channels, kernel size, stride size, and group size for layer p, respectively. Equation (4) represents the TPC score for a linear layer, which is equivalent to its output dimension. Finally, for a transformer layer shown in Figure 4, the TPC score is Equation (5), where A, D, and r is the Q-K-V dimension, embedding dimension, and MLP ratio.

$$O_p = \frac{C_{o,p} * {k_p}^2}{{s_p}^2 * g_p}, \tag{3}$$

$$O_p = C_{o,p}, \tag{4}$$

$$O_p = (A \times 3) \times D \times (D \times r) \times D, \tag{5}$$

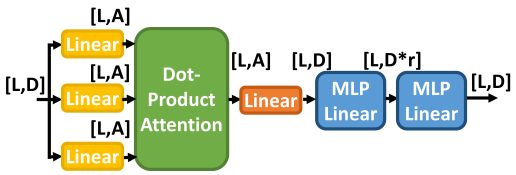

Fig. 4. Overview of Transformer layer architecture.

However, as neural networks become deeper and larger, the original TPC score can become extremely large. Therefore, [9] takes the logarithm of each layer's TPC score for easier observation and usage. This transforms the multiplication of

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scores into an addition, resulting in a smaller and more manageable TPC score, as shown in Equation (6)

$$S_t = \sum_p \log(O_p). \tag{6}$$

We also investigated the relationship between the TPC score and model expressivity under MIM pre-training. For consistency, we used the same 32 ViT models as those in the Entropy score correlation analysis. Figure 5 and Table II highlight a strong correlation between the TPC score and Top-1 accuracy on the ImageNet-100 dataset, further demonstrating the effectiveness of the TPC score as a measure of model performance.

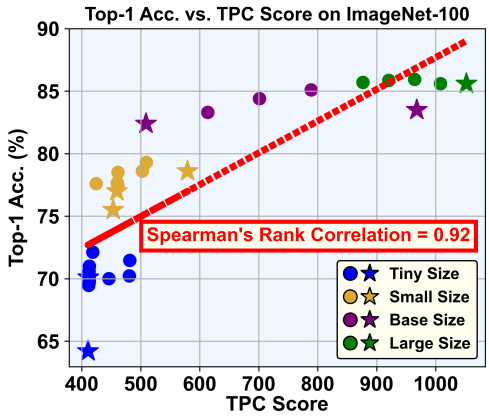

Fig. 5. Relationship between the TPC score and low-labeled data accuracy along with Spearman's rank correlation coefficient.

TABLE II
RELATIONSHIP BETWEEN TPC SCORES AND TOP-1 ACCURACY FOR DIFFERENT VIT MODELS ON IMAGENET-100 10% LABELED DATA.

Model	TPC Score	Top-1 Acc. (%)
DeiT-T	410.7	64.2
Autoformer-T	411.2	70.1
ResNet-50	453.3	75.5
DeiT-S	459.6	77.0
MCMAE-S	579.3	79.7
DeiT-B	508.9	82.4
Swin-B	968.0	83.5
DeiT-L	1052.0	85.6

C. GEN-TPC-NAS Algorithms

GEN-TPC-NAS combines Entropy score and TPC score with genetic algorithms. As illustrated in Algorithm 1 and Figure 6, it begins with a population P of size N, initializing each individual in the population with random architecture parameters. Then it does Mutation: randomly select individuals from P and apply mutation to generate new individuals or Crossover: Use crossover techniques by randomly selecting two individuals from the population and exchanging their architecture parameters to create new individuals. If the newly generated individuals don't meet the constraints of parameter

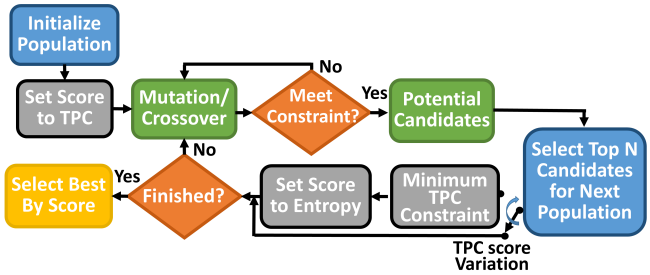

Fig. 6. Overview of the search algorithm in GEN-TPC-NAS.

count or computational complexity, discard these individuals and generate new ones. After N valid individuals are created, we combine the new generation with the previous generation's individuals and select the top N individuals with the highest TPC scores to form the next generation. Suppose the difference between the maximum and minimum TPC scores among the N individuals in population P is smaller than a predefined threshold T_0. In that case, it indicates that the N candidates within the population exhibit no significant differences in their TPC scores. At this point, the evaluation score is switched from TPC score to Entropy score to assess the generalization ability of the architectures. Additionally, a minimum TPC Score is introduced as a new constraint during the search process, signifying that the search aims to discover architectures with the highest generalization capability, provided their expressivity exceeds a certain value. After sufficient generations, the individual with the highest evaluation score in the final population represents the best model architecture.

This approach maintains the fast search advantage of zero-shot NAS by only using model architecture parameters and one forward propagation, thus avoiding the need for extensive training and keeping search time less than one hour.

D. Search Space Design

We take inspiration from [29] and design a search space that includes four adjustable factors within Transformer building blocks: embedding dimension, Q-K-V dimension, head dimension, and MLP ratio. The number of heads in the multi-head attention module is determined by dividing the Q-K-V dimension by the head dimension. Details of the search space configurations for various Transformer architectures tailored to CV and NLP tasks are provided in Table III. Notably, we define the search range for the Q-K-V dimension to be slightly larger than that of the embedding dimension, allowing for greater flexibility in the number of attention heads. This design enables each attention module to dynamically adjust and determine the optimal number of heads as needed.

The constraints applied during the search process primarily include the parameter count, computational complexity, and the number of layers. While having more parameters in an architecture can potentially improve both training and test

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Algorithm 1 GEN-TPC-NAS Algorithm

Require: search space $\mathcal{S}$, constraints K, maximal depth D, number of iterations M, population size N, initial structure F_0, difference threshold T_0

1: Initial population $\mathcal{P}(F_0)$
2: $switchFlag \leftarrow False$
3: $TPC_{Threshold} \leftarrow 0$
4: **for** $m = 1, 2, \ldots, M$ **do**
5: Randomly select two models F_1, F_2 from the population.
6: **if** mutation condition is met **then**
7: Structure $\hat{F} = \text{MUTATE}\,(F_1, \mathcal{S})$
8: **else**
9: Structure $\hat{F} = \text{CROSSOVER}\,(F_1, F_2, \mathcal{S})$
10: **end if**
11: **if** $\hat{F}$ does not satisfy hardware constraints K or has more than D layers **then**
12: Go to line 6
13: **else**
14: Calculate $S_t(\hat{F})$
15: **if** $switchFlag = False$ **then**
16: $S \leftarrow S_t$
17: Insert $\hat{F}$ to population $\mathcal{P}$
18: **else**
19: $S \leftarrow S_e$
20: **if** $S_t(\hat{F}) \leq TPC_{Threshold}$ **then**
21: Go to line 6
22: **else**
23: Calculate $S_e(\hat{F})$
24: Insert $\hat{F}$ to population $\mathcal{P}$
25: **end if**
26: **end if**
27: **end if**
28: Remove F_{min}, the model with the lowest S, if $|\mathcal{P}|$ exceeds N
29: **if** $switchFlag = False$ and $S_t(F_{\text{max}}) - S_t(F_{\text{min}}) \leq T_0$ **then**
30: $TPC_{Threshold} \leftarrow S_t(F_{\text{min}})$
31: $switchFlag \leftarrow True$
32: **end if**
33: **end for**

Return F_{max}, the model with the highest S

performance, it is often difficult to determine whether the performance gain stems from the architecture itself or simply from the increase in parameter count. Furthermore, model deployment typically imposes limits on parameter size and computational complexity. Therefore, setting constraints on these factors is both reasonable and essential to ensure that the searched models are practical for real-world implementation. Regarding the constraint on the model's number of layers, it is evident from Equation (2) and Equation (6) that both the Entropy and TPC scores can increase indefinitely as the number of layers grows. However, if the parameter count and computational complexity are fixed, adding more layers is unlikely to result in significant performance improvements. Therefore, we impose a limit on the number of layers during the search process.

TABLE III
SEARCH SPACE DESIGN OF VITS AND GPT2. TUPLES OF THREE VALUES IN PARENTHESES REPRESENT THE LOWEST VALUE, HIGHEST, AND STEPS.

Search Space	DeiT-Tiny	DeiT-Small	GPT2
Embed Dim	(168, 216, 24)	(384,448,32)	(768,832,32)
Q-K-V Dim	(136, 288, 32)	(352,448,32)	(736,832,32)
Head Dim	(32,96,32)	(32,96,32)	(32,96,32)
MLP Ratio	(2.0,4.0,0.125)	(2.0,4.0,0.125)	(2.0,4.0,0.125)

IV. EXPERIMENTS

In this section, we first provide the experimental settings of the proposed GEN-TPC-NAS. Then we compare GEN-TPC-NAS with the state-of-the-art models. Finally, we conduct ablation studies on ViT models and GPT2. All the experiments are conducted on 2 GeForce RTX 4090 GPUs.

A. Experimental settings

Dataset Description. For the computer vision task, we evaluate the effectiveness of the proposed method on the large-scale ImageNet-1k [6] image classification and ADE20K [36] semantic segmentation. Additionally, we use ImageNet-100, comprising the first 100 classes from ImageNet-1k, for ablation studies. For the NLP task, we utilize the Cleaned Common Crawl dataset and the Cleaned Wikipedia dataset as pre-training datasets. The test dataset used for fine-tuning is WikiText-2, a curated collection of over 100 million tokens extracted from verified Wikipedia articles.

Implementation Details. During the search phase, we maintain a fixed population size of 100 and iterate through 1000 generations. The difference threshold T_0 is set to 0.5. For the computer vision task, we select DeiT-Tiny as the baseline model for the challenging ImageNet-1k Experiments. The design of the search space is detailed in Table III, while the search constraints are determined by the original architecture's parameter size, computational cost, and depth.

In the pre-training stage, we employ the MaskTAS algorithm [25] and train for 100 epochs. We use a batch size of 1024 and a base learning rate of 1.5e-4. During fine-tuning, the entire dataset is used to train 200 epochs for the standard ImageNet-1k benchmark. For the low-labeled data ImageNet-1k benchmark, we fine-tune models for 200 epochs on 20–40% of ImageNet-1k labeled data to evaluate model performance under limited annotation conditions. For ADE20K, we follow the same setting in MAE and adopt UperNet [37] as our framework with a GEN-TPC-Tiny pre-trained backbone. The input image resolution is 512×512 for training and evaluating. We use mIoU as the evaluation metric.

For the NLP task, we adopt a 12-layer GPT-2 model with 110 million parameters as the baseline. The search space design is shown in Table III, with constraints ensuring similar computational and parameter sizes. We set the learning rate to 6e-4 and the batch size to 6. We pre-train GPT2 with an

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TABLE IV
COMPARISON WITH STATE-OF-THE-ART VITS ON IMAGENET-1K.

Method	FLOPs (G)	Params. (M)	Top-1 Accuracy (%)	Design Type	Search Cost (GPU Days)
DeiT [30]	1.2	5.7	72.2	Manual	-
Autoformer [29]	1.3	5.7	74.7	One-Shot	24
ViTAS-C [31]	1.6	5.6	74.7	One-Shot	32
GraSP [32]	1.3	5.5	74.3	Zero-Shot	0.4
GradSign [33]	1.5	6.0	74.9	Zero-Shot	1.7
MGM [34]	1.5	6.0	75.0	Zero-Shot	0.14
Grad-norm [35]	1.4	5.9	74.8	Zero-Shot	0.03
TF-TAS [4]	1.4	5.9	75.3	Zero-Shot	0.5
T-Razor [5]	1.4	5.9	75.5	Zero-Shot	0.4
MaskTAS [25]	1.3	5.8	75.6	One-Shot+SSL	N/A
GEN-TPC-NAS	1.3	5.4	**77.5**	Zero-Shot+SSL	0.02

input sequence length of 1024 for 100000 steps. Fine-tuning is performed with a learning rate of 1e-5 for 4 epochs. We use 100% and 0-10% labeled data of the Wikitext-2 dataset for the full and low-labeled data experiment, respectively. The evaluation metric is perplexity. Lower perplexity indicates better model performance and the ability to generate sentences closer to the original.

B. Results

Vision Transformers (ViTs). We first compare GEN-TPC-NAS with previous NAS methods on the standard ImageNet-1K benchmark. As shown in Table IV, our GEN-TPC-NAS achieves an accuracy of 77.5%, surpassing the performance of previous state-of-the-art models. This highlights the combination of NAS and SSL could learn more useful representations than traditional supervised learning. Furthermore, our comparison with MaskTAS [25] demonstrates that the combination of zero-shot NAS and SSL outperforms the self-supervised one-shot NAS framework. For the low-labeled data ImageNet-1k benchmark, we compare our proposed method with TF-TAS [4] and MaskTAS [25]. Using the architectures reported in their papers, we apply the same pre-training and fine-tuning settings. The experimental results in Figure 7 demonstrate that GEN-TPC-NAS achieves higher accuracy with 20-40% labeled data. This indicates that our method exhibits better generalization ability compared to other approaches in scenarios with limited annotations. Furthermore, compared to the baseline DeiT-Tiny trained in a supervised-learning manner, GEN-TPC-NAS can reduce the required labeled data by up to 75% while maintaining the same level of accuracy.

We also evaluate the generalization ability of the proposed method by transferring the searched architectures to a semantic segmentation task. The results shown in Table V indicate the proposed GEN-TPC-NAS can still achieve better performance than the other methods.

GPT2. Table VI compares the performance of the baseline GPT2, TPC-GPT2, and GEN-TPC-GPT2. The proposed GEN-TPC-NAS achieves a further 2.9% reduction in perplexity compared to TPC-NAS, demonstrating its superior performance. This result validates that GEN-TPC-NAS is not only applicable to image classification tasks based on ViT but also effective in identifying high-performing models for text generation tasks.

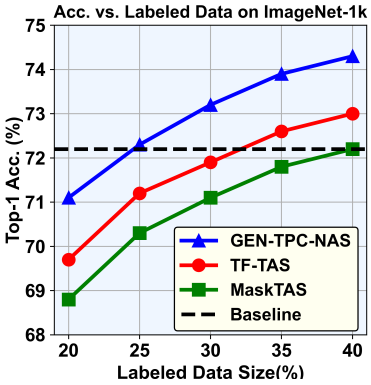

Fig. 7. Comparison with other NAS methods on low-labeled data ImageNet-1k benchmark.

TABLE V
PERFORMANCE COMPARISON (MIOU) ON ADE20K SEMANTIC SEGMENTATION.

Model	Pre-train Data	mIOU
DeiT-Tiny [30]	IN1K w/ labels	38.0
MAE-Tiny [8]	IN1K w/o labels	37.6
MaskTAS-Tiny [25]	IN1K w/o labels	40.6
GEN-TPC-Tiny	IN1K w/o labels	**41.2**

TABLE VI
COMPARISON BETWEEN ARCHITECTURE DISCOVERED BY GEN-TPC-NAS, TPC-NAS AND BASELINE GPT2.

Model	FLOPs (G)	Params. (M)	TPC Score	Entropy Score	Perplexity
GPT2 [18]	108	84.9	503.1	127.4	21.1
TPC-GPT2 [9]	108	84.7	504.3	129.8	13.5
GEN-TPC-GPT2	**104**	**84.9**	**504.2**	**131.0**	**10.6**

For the low-labeled data Wikitext-2 experiment, we use 0-10% labeled data to fine-tune our GEN-TPC-GPT2. The comparison results with baseline GPT2 and TPC-GPT2 are shown in Figure 8. GEN-TPC-GPT2 can achieve baseline-level perplexity by less than 1% of labeled data. In comparison to TPC-GPT2, our model can attain the same perplexity using only 3% labeled data, reducing the number of labels by 97%.

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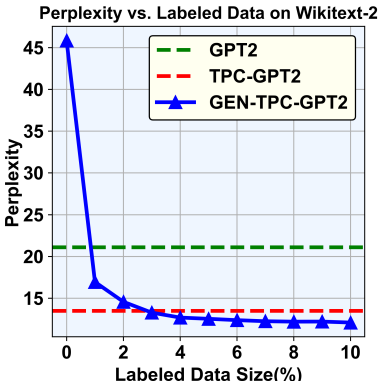

Fig. 8. Wikitext-2 Results on GPT2-series.

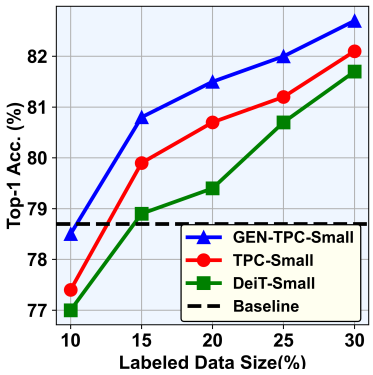

Fig. 9. low-labeled data ImageNet-100 Results on Small size of ViTs.

This demonstrates the high generalization ability of our model in the field of NLP tasks.

C. Ablation Studies

Experiments on ViT Models of Different Sizes. To further validate the generalization ability of our method across different parameter scales, we conducted experiments using DeiT-Small and DeiT-Tiny on the ImageNet-100 dataset. The results in Figure 9 and Figure 10 demonstrate that the architectures identified by GEN-TPC-NAS achieve accuracy comparable to that of the baseline models trained with supervised learning, even when utilizing only 10% of the labeled data. When comparing the amount of labeled data required for the original models and GEN-TPC-NAS to achieve the same baseline accuracy, we observe that the Small-series reduces the required labeled data by 5%, while the Tiny-series reduces it by an additional 15%.

TABLE VII
COMPARISON BETWEEN ARCHITECTURE DISCOVERED BY GEN-TPC-NAS, TPC-NAS AND BASELINE VITS.

Model	FLOPs (G)	Params. (M)	TPC Score	Entropy Score	Acc.@30% label (%)
DeiT-Small [30]	4.7	22.1	459.6	132.1	81.7
TPC-Small [9]	4.6	21.1	461.4	133.9	82.1
GEN-TPC-Small	**4.6**	**20.8**	**461.4**	**135.3**	**82.7**
DeiT-Tiny [30]	1.3	5.7	410.7	117.4	72.6
TPC-Tiny [9]	1.3	5.4	412.5	117.6	77.0
GEN-TPC-Tiny	**1.3**	**5.4**	**413.0**	**120.2**	**77.9**

Effect of TPC Score Difference Threshold. As shown in Table VIII and Table IX, we observed that when TPC Score difference threshold $T_0 = 0.5$, the GEN-TPC-NAS identified architectures balancing both TPC and Entropy scores, resulting in the best performance. When $T_0 = 0$, the search essentially becomes equivalent to TPC-NAS, focusing solely on evaluating architectures based on the TPC score. On the other hand, when $T_0 = 1$, the search algorithm prioritizes generalization before identifying architectures with sufficiently high TPC Scores. As a result, the selected architectures exhibit higher Entropy scores but remain sub-optimal due to a gap in TPC scores compared to the optimal architecture.

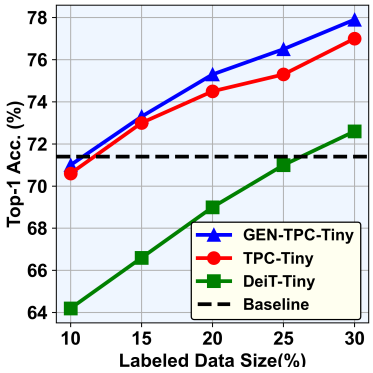

Fig. 10. low-labeled data ImageNet-100 Results on Tiny size of ViTs.

TABLE VIII
EFFECT OF TPC SCORE DIFFERENCE THRESHOLD FOR VITS.

Setting	TPC Score	Entropy Score	Acc.@10% label (%)	Acc.@30% label (%)
GEN-TPC-Small, T_0=0	461.4	133.9	77.4	82.1
GEN-TPC-Small, T_0=0.5	461.4	135.3	**78.5**	**82.7**
GEN-TPC-Small, T_0=1.0	460.3	135.3	77.7	82.0
GEN-TPC-Tiny, T_0=0	412.5	117.6	70.6	77.0
GEN-TPC-Tiny, T_0=0.5	413.0	120.2	**71.0**	**77.9**
GEN-TPC-Tiny, T_0=1.0	412.1	120.4	70.8	76.9

TABLE IX
EFFECT OF TPC SCORE DIFFERENCE THRESHOLD FOR GPT2.

Setting	TPC Score	Entropy Score	Perplexity @1% label	Perplexity @3% label
GEN-TPC-GPT2, T_0=0	504.3	129.8	17.7	14.9
GEN-TPC-GPT2, T_0=0.5	504.2	131.0	**16.9**	**13.3**
GEN-TPC-GPT2, T_0=1.0	503.1	131.0	17.2	13.5

V. CONCLUSION

In this study, we introduce GEN-TPC-NAS, a novel self-supervised zero-shot neural architecture search (NAS) method for Transformer architectures. By combining two zero-shot proxies from complementary perspectives, our method identifies architectures that effectively balance expressivity and generalization. The integration of NAS and self-supervised learn-

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ing (SSL) not only significantly reduces the annotation costs of supervised learning, but also facilitates the efficient training of transformers. Experimental results on image classification and text generation tasks demonstrate that GEN-TPC-NAS can achieve state-of-the-art performance in both standard and low-labeled data scenarios. These findings highlight the potential of GEN-TPC-NAS to automate the neural architecture design process for real-world applications, reducing dependency on human expertise and lowering data labeling costs.

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